



Study on microwave coherent power combining based on 20-kW S-band CW power-adjustable magnetrons

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ABSTRACT

In order to meet the requirements of large-power capacity in microwave-energy industry, the microwave coherent power combining by multiple magnetrons is a valid method to gain a large-power and high efficiency microwave source. In microwave energy applications, microwave absorption of industrial materials is varying in the heating duration and dependent on the dynamic change of materials. It is the first time that a 20-kW continuously power-adjustable injection-locked magnetron source has been achieved, and according to the experiment results, the maximum microwave coherent power combining output reaches 33.1 kW, and the coherent power combining efficiency is higher than 93%. The experiments show that the unbalanced output power between the two magnetrons does not affect greatly on the microwave coherent power combining efficiency. It is a great progress to develop the microwave power source which satisfies the power capacity requirement of microwave energy industry with the real-time power adjustment for the microwave industrial applications.

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1. Introduction

With the increasing demands for clean energy, substantive microwave industrial applications require high power capacity, such as microwave plasma diamond film deposition [1,2]. Therefore, microwave coherent power combining technology has become an effective solution [3,4]. In order to realize multiple magnetrons coherent power combining, it is necessary to solve the problem of frequency and phase instability of magnetron output [5]. In addition, in the process of the microwave heating, the capacity of industrial materials microwave absorption is dynamic changing, if the microwave source cannot control the power on time, it will lead to the decline in the utilization of microwave sources, and even the risk of material burn or explosion [6,7]. Real-time adjustment of high-power microwave source is the practical requirement of industrial application. Therefore, it is crucial to improve the utilization rate of microwave source by developing the microwave power source, which satisfies the power capacity requirement of microwave energy industry with the real-time power adjustment.

There have been some researches on the injection locking and power combining of CW (continuous wave) magnetron in the SSPS (space solar power satellite) or accelerator applications [8,9]. In 2004, Shinohara et al. used a commercial microwave oven magnetron to achieve a coherent power combining source of 300–500 W, which is phase- and power-control power combining source. This is the first time for a continuous wave magnetron that is controlled by the phase and amplitude [10,11]. Through the phase control of the injection-locked magnetron, our previous work achieves the coherent power combining of two fixed output injection-locked magnetrons, and the average combining efficiency is about 96% [12]. In order to realize the power and phase control of the high power magnetron sources at the same time, the improvement of magnetrons output power adjusting is necessary.

In this paper, the adjustment on water-cooling system and high-voltage power supply control of CK2091 fixed output power large-power magnetron is applied to improve the output power of the magnetron, and realize the injection locking and power adjustment of the magnetron. Due to magnetron frequency pushing, magnetron power-control will affect the frequency range of the injection locking, resulting in an injection locking frequency range shift in different output power [13]. Therefore, this paper presents the effect of magnetron output power adjustment on its injection phase locking and studies the effect of power combining on the coherence power combining efficiency of two magnetrons with different power.

2. Research on power-adjustable and injection locking of magnetron

In experiment, we used the magnetron manufactured by Nanjing Sanle Microwave Technology Co., Ltd. to exposit the effect of the magnetron power-control on the injection-locked frequency range of the magnetron. The DC negative high power is provided by switched-mode power supply, whose anode current and filament current is adjustable. The DC voltage is -11.5 V and the magnetic field is 1550 Gs in the experiments. With the change of anode current, the DC high voltage remains almost constant. Then current of electromagnet is kept at 3.1 A to maintain the constant magnetic field.

2.1. Power-adjustment of 20-kW S-band CW magnetron

Figure 1 shows the frequency spectra of different microwave output power signal of a free oscillation magnetron system. The detailed input and output parameters for magnetron in Figure 1 is shown in Table 1. In the experiment, the anode current is changed by step of 0.01 A. When the magnetron microwave output increases, the filament current should be reduced to ensure the stability of magnetron. According to the experimental measurements of the frequency spectrum and output power, significant magnetron frequency pushing effect has been observed [14]. When the anode current of the magnetron increases, the microwave output power increases and the center frequency of the free oscillation magnetron increases gradually.

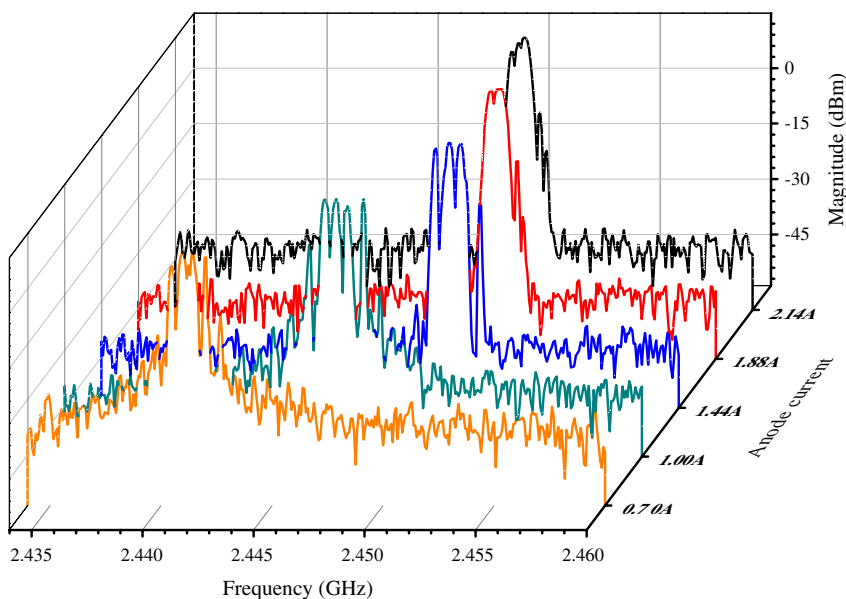


Figure 1. Spectra of experimental measurements at various operating currents in free oscillation.

Table 1. The detailed parameters of injection-locked magnetron at various output.

Anode current (A)	Central frequency (GHz)	Filament current (A)	Output power (kW)
0.70	2.442	40.5	5.74
1.00	2.445	37.4	8.69
1.44	2.449	33.0	11.62
1.88	2.450	28.5	15.20
2.14	2.450	26.2	18.90

Table 2. Magnetron injection-locked experiments measurement.

Injected signal power (W)	Output power (kW)	Phase-locking bandwidth scope (GHz)	Injection-locked bandwidth (MHz)
140	5.7	2.4342–2.4472	13.0
140	8.7	2.4418–2.4495	7.7
140	11.6	2.4478–2.4506	2.8
140	15.2	2.4482–2.4516	3.4
140	18.9	2.4480–2.4509	2.9

2.2. Bandwidth of injection-locked magnetron under power-control

After the magnetron power-control accomplishment, it is necessary to injection-lock the magnetron and study injection-locking frequency range whether or not meets the demand of coherent power combining after magnetron power-control. Table 2 shows the injection-locked frequency range and bandwidth with the magnetron in different microwave output under the same injected signal power. According to Adler condition, the locking bandwidth is related to the quality factor of the magnetron, the injection ratio, and the oscillation frequency [1]. As a consequence of that, the shift of center frequency of magnetron oscillation will affect the injection-locked bandwidth. It can be seen from the experiment

results, the center frequency of free oscillation and injection-locked frequency range is changed with the increase of magnetron microwave output power.

3. Coherent power combining experiment of two magnetrons

Figure 2 shows a block diagram of the microwave power coherent combining experiment based on two S-band 20 kW CW power-adjustable injection-locked magnetrons. The red arrow line represents the magnetron's power supply system and the path of large microwave power. The two magnetrons are powered by switched-mode power supply. The negative high DC voltage, electromagnetic field current, filament voltage and operating current can be controlled by the host computer. The purple line represents the main component of the phase control of microwave combining system. This part is based on MMIC chip AD8302 phase detector to detect phase, providing 0–180° phase difference detection range and the sensitivity of 10 mV/degree. The phase shifter uses an analog phase shifter with USB port to adjust the phase of magnetron two. The minimum adjustment accuracy of the phase shifter

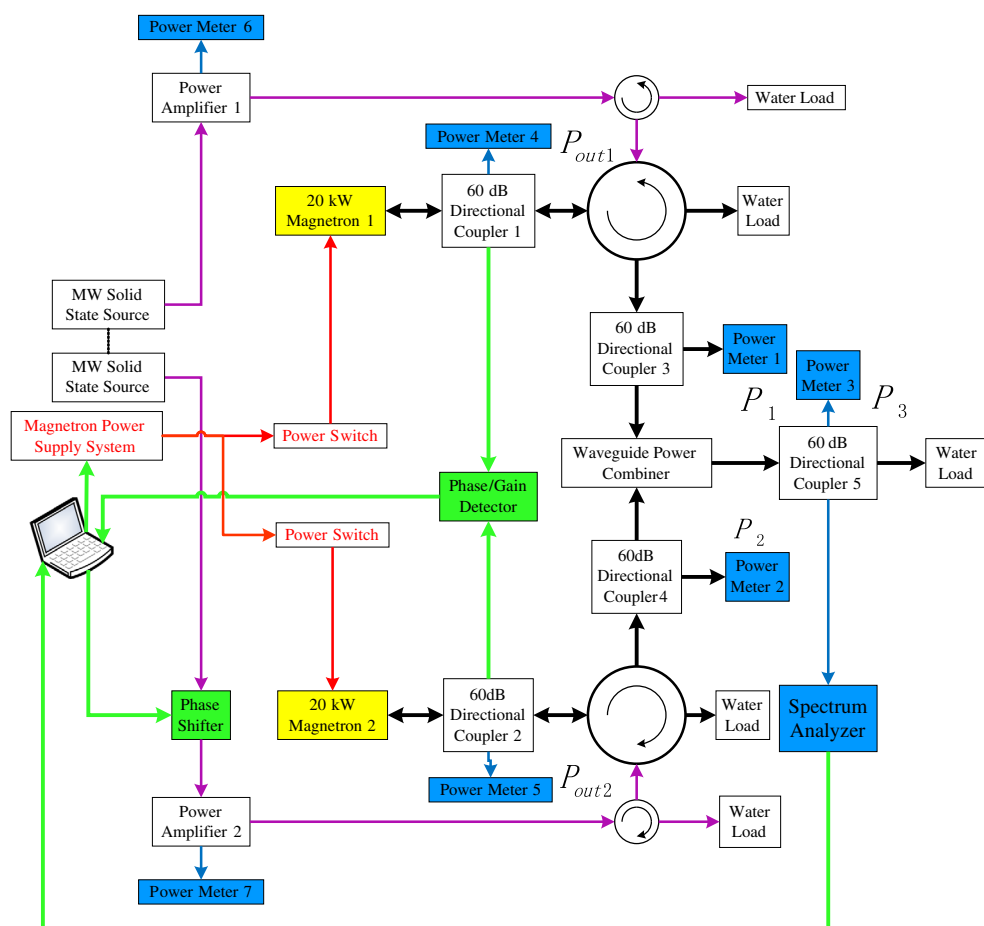


Figure 2. Schematic of microwave power combining system based on two-way injection-locked 20-kW CW power-adjustable magnetrons.

is 0.25° and the adjustment range is 360° . Two four-port circulators are used for the isolation and protection between injected signal and magnetron output.

In large-power microwave applications, the circulator attenuation loss is large, and the attenuation is different when the frequency is different. The power meters one and two are used to measure the input power P_1 and P_2 at the two input ports of the power combiner, respectively. The power meter three is used to measure the output power P_3 of the microwave power combining. The power meters four and five are used to measure the actual output power P_{out1} and P_{out2} of the two magnetrons. The power meters six and seven are used to measure the microwave power of the injected signals of the two magnetrons, respectively. The frequency spectrum analyzer is connected to the directional coupler five to monitor the spectral characteristics combining process. An H-plane waveguide Tee power combiner is used to achieve coherent combining of the output power of two magnetrons. Figure 3 shows the experimental system.

4. Coherent power combining experiment results

4.1. Coherent power combining experiments with balanced input power

Due to the shift of the injection-locked frequency range of the magnetron, coherent power combining is possible only when the magnetrons are locked at same frequency point. When the output power of the magnetron is too low, the output spectrum of the magnetron fluctuates greatly, and then it is difficult to realize injection-locked and the coherent power combining. Table 3 shows the measured results of coherent power combining experiments when the power ratio is close to one (the power of two magnetrons is essentially the same). The coherent power combining efficiency is more than 93% when the two magnetrons have same frequency, amplitude and the phase difference is small.



Figure 3. Microwave power combining system based on two injection-locked 20-kW CW power-adjustable magnetrons.

Table 3. Injection-locked experiments with balanced input power.

No.	P_1 (kW)	P_2 (kW)	Frequency (MHz)	P_3 (kW)	Efficiency
1	5.3	5.37	2440.1	10.1	95.6%
2	10.1	9.7	2449.3	18.6	93.9%
3	14.7	14.3	2449.4	27.8	95.8%
4	17.7	17	2450.0	33.1	95.3%

Table 4. Power combining efficiency variation due to unbalanced input power.

No.	P_1 (kW)	P_2 (kW)	Frequency (MHz)	k	P_3 (kW)	Efficiency
1	10	10.1	2449.3	1.00	19.1	95.0%
2	10	11.0	2449.3	1.10	19.7	93.8%
3	10	12.5	2449.4	1.25	21.3	94.6%
4	10	13.1	2449.4	1.31	21.9	94.8%
5	10	13.7	2449.4	1.37	22.4	94.5%
6	10	14.9	2449.4	1.49	23.5	94.3%
7	10	16.0	2450.0	1.60	24.3	93.4%

The coherent power combining efficiency shown in Table 3 is slightly lower than the experimental efficiency of 96% in our previous experiments in reference [12] with adjustable microwave output. The change of coherent power combining efficiency is mainly due to the improvement of the magnetron high voltage power supply and the change of ripple on the phase stability of the injection lock. In addition, the loss and match of waveguide Tee power combiner is changed at different frequencies.

4.2. Experiments with unbalanced input power

In the combining experiment, the microwave output power-control has two power imbalance situations. By adjusting the microwave output power of a magnetron, the influence of the output power ratio on the coherent power combining efficiency of two magnetrons is studied. The magnetron one output power P_1 is maintained at 10 kW before injecting a signal. The microwave output power P_2 of the magnetron two is adjusted to 10.1, 11, 12.5, 13.1, 13.7, 14.9, and 16 kW. In different microwave output power, the injected signal frequency settings need to be adjusted according to the power to ensure that the two magnetrons are locked at the same frequency point.

After two magnetrons being injection-locked at the same frequency, the optimum efficiency of the coherent power combining is found by adjusting the phase of the magnetron two as shown in Table 4. The experimental results show that when the output power of one magnetron maintains the same, the power value of the other magnetron is changed from 10.1 to 16 kW. In other word, the power ratio changes from 1 to 1.6, the change of coherent combining efficiency is not obvious. It can be seen that it is more important keep the phase coincidence of the two magnetrons in the coherent power combining experiment to achieve a higher combining efficiency. In the power combining process of multiple magnetrons, the appropriate adjustment of power does not have a significant impact on the coherent power combining efficiency.

5. Conclusion

In this paper, we enhance the control program of magnetron and high-voltage power supply and other aspects, and improve the fixed output power of the magnetron into microwave output power-adjustable magnetron. A 20-kW injection-locked power-control magnetron is achieved and applied to coherent power combining for the first time with a maximum combining output power of 33.1 kW and the combining efficiency more than 93%. Through the experimental study, the frequency pushing effect of the adjustable magnetron is found, and its influence on the injection-locked frequency range of the magnetron is analyzed. The effect of coherence power combining efficiency was studied in the two cases of power balance and unbalance of two magnetron systems. The experimental results show that the power ratio has little effect on the efficiency of coherent power combining when the microwave power of the two magnetrons is different. Therefore, in the case of power adjustment, large-power microwave source based on power-control injection-locked magnetron can still ensure high power combining efficiency.

Disclosure statement

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